Population-level Corpus-Derived Age-of-Acquisition Estimates for Negative Morphemes in Six Languages

Overview – Children begin acquiring negative morphemes like *no* early, though exactly when remains contested. Recent experimental evidence suggests that 18-to–24-month-olds can already understand negative sentences in English (de Carvalho, & Dautriche, 2025), French (de Carvalho et al., 2020), and Hungarian (Szabó & Kovács, 2025); yet other studies have failed to find such comprehension abilities in the same age range, or even in older children (Austin et al., 2014; Feiman et al., 2017; Nordmeyer & Frank, 2014; Reuter et al., 2018), suggesting possible experimental task effects. Children's early production of negative morphemes has been documented across languages and corpora (Bloom, 1970; Choi, 1988; Pea, 1980; Cameron-Faulkner et al., 2007; Jasbi et al., 2020; inter alia), but no systematic method exists for estimating the population-level age at which children begin producing negative morphemes using corpus data. We address this gap by applying Bayesian growth-curve modeling to child language corpora, developing a novel approach to estimating population-level age of acquisition for negative morphemes. Our results show that across six languages (English, Spanish, French, Mandarin Chinese, German, and Japanese), children begin producing a wide variety of negative morphemes between 12 and 24 months. Together with the experimental evidence cited above, these results provide converging support for the view that negation emerges in both comprehension and production within children's second year of life.

Methods – We used the following corpora in CHILDES CHILDES (MacWhinney 2000), accessed via the online database childes-db.stanford.edu and its associated R package {childsr} (Sanchez et al. 2019): [list the collections] Our collection contained [token and child count per language]. We compiled and edited a list of negation words in . All tokens were annotated for the age of the child (in months) as well as whether the token was produced by parents or children. We computed the "cumulative relative frequency" of each function word by dividing its frequency up to that month by the frequency of all the words produced by children/parents up to that month.

Results – Figure 1 shows the cumulative relative frequency of X negative morphemes by children's age (months) in English, French, German, Mandarin Chinese, Spanish, and Japanese. For most negative morphemes, cumulative relative frequencies follow a nonlinear S curve across development with a period of accelerated growth, a period of slowdown, and a final stage of stable production. We fit separate Bayesian growth curve models to the cumulative relative frequency of each function word using the BRMS package (Bürkner 2017). We used growth curves with three parameters: 1. Lambda (lag): estimating the age at which production starts; 2. Miu: maximum rate of growth; and 3. A: maximum growth (Figure 2, right). Uniform priors were chosen for all three parameters. Figure 2 (Left) shows the onset of production estimates for all negative morphemes in our study. ($\beta = =$, $t = =$, $p = =$).

Word Count: 497

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